

Locara Atlas MVP Architecture & Schema

This document outlines the architecture, technology stack, and database structures for the Locara Atlas MVP.

Overview

The Locara Atlas MVP is designed as a modern, high-performance web application tailored for delivering enterprise video datasets securely. It replaces manual Google Drive sharing with a premium, brand-aligned client portal.

Technology Stack

1. **Frontend Framework:** Next.js 15 (App Router) with TypeScript for type-safe, server-rendered React.
2. **Styling:** Tailwind CSS + Shadcn UI + Base UI for rapid, highly-customizable and accessible design tokens.
3. **Animations:** Framer Motion for premium micro-interactions and page transitions.
4. **Backend & Database:** Supabase (PostgreSQL) handling Auth, Database, and Row-Level Security (RLS).
5. **Video Storage & Streaming:** Videos hosted on Cloudflare R2 / AWS S3 and streamed natively via HTML5 `video` tags to reduce application weight.
6. **Hosting:** Vercel for CI/CD and edge deployment.

Supabase PostgreSQL Schema

The database is built on PostgreSQL via Supabase, with Row-Level Security (RLS) strictly enforced.

1. organizations

Represents client companies.

- `id` : uuid (PK)
- `name` : text NOT NULL
- `logo_url` : text
- `is_active` : boolean DEFAULT true
- `created_at` : timestampz

2. users

Locara Admins and Client Users.

- `id` : uuid (PK) -> Matches Supabase Auth
- `organization_id` : uuid FK -> `organizations`
- `name` : text NOT NULL
- `email` : text UNIQUE NOT NULL
- `role` : text CHECK IN ('admin', 'client')
- `is_active` : boolean DEFAULT true
- `created_at` : timestampz

3. collections

Named groups of videos assigned to a specific organization.

- `id` : uuid (PK)
- `organization_id` : uuid FK -> `organizations`
- `title` : text NOT NULL
- `description` : text
- `cover_image_url` : text

- `tags : text[]`
- `is_published : boolean DEFAULT false`
- `created_at : timestampz`

4. videos

The core dataset records containing metadata and the streaming URL.

- `id : uuid (PK)`
- `collection_id : uuid FK -> collections`
- `video_id : text NOT NULL`
- `worker_id : text`
- `task_type : text[]`
- `normalized_tags : text[]`
- `duration_minutes : integer`
- `video_length : text`
- `environment : text`
- `recording_date : date`
- `video_url : text NOT NULL`
- `thumbnail_url : text`
- `file_size : text`
- `resolution : text`
- `frame_rate : text`
- `audio_quality : text`
- `hands_visible : boolean`
- `lighting_quality : text`
- `pii_check_status : text`
- `qa_status : text DEFAULT 'Pending'`
- `created_at : timestampz`

5. dataset_requests

Requests submitted by clients for more data.

- `id : uuid (PK)`
- `organization_id : uuid FK -> organizations`
- `user_id : uuid FK -> users`
- `task_type : text`
- `environment : text`
- `hours_needed : integer`
- `deadline : date`
- `notes : text`
- `status : text DEFAULT 'Submitted'`
- `created_at : timestampz`

Row-Level Security (RLS) Policies

- **Collections:** Clients can SELECT only where `organization_id` matches their own.
- **Videos:** Clients can SELECT videos whose `collection_id` belongs to an organization they have access to.
- **Dataset Requests:** Clients can INSERT new requests and SELECT only their own organization's requests. Admins have global access.